

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (IT)

May 2012

IT304 Modeling Simulation & Operational Research

Date: 04.05.2012, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q 1 (a) State whether following statements are true or false [04]

- 1 The feasible region of an LP problem is always bounded.
- 2 The variables must be nonnegative in a standard LP problem.
- 3 The optimum solution of an LP problem cannot contain slack variables in the basis.
- 4 The slack and surplus variables can be unrestricted in sign.

(b) The following table summarizes the key facts about two products, A and B, and the resources, Q, R and S, require to produce them, [03]

Resource	Resource Usage per Unit Produced		Amount of Resource Available
	Product A	Product B	
Q	2	1	2
R	1	2	2
S	3	3	4
Profit per unit	3	2	

All the assumptions of linear programming hold. Formulate a linear programming model for this problem.

Q 2 (a) Convert the following primal problem into dual problem. [05]

$$\text{Maximise } Z = 2x_1 + 2x_2 + 4x_3$$

subject to the constraints

$$2x_1 + 3x_2 + 5x_3 \geq 2$$

$$3x_1 + x_2 + 7x_3 \leq 3$$

$$x_1 + 4x_2 + 6x_3 \leq 5$$

$$x_1, x_2, x_3 \geq 0$$

(b) What are main characteristics of Operations research? State any four applications of Operations Research? [04]

- (c) Solve the following LPP using graphical method

[05]

$$\text{Maximize } z = 5x_1 + 4x_2$$

$$\text{Subject to, } 6x_1 + 4x_2 \leq 24, x_1 + 2x_2 \leq 6, -x_1 + x_2 \leq 1, x_2 \leq 2$$

$$\text{and } x_1 \geq 0, x_2 \geq 0$$

OR

- Q 2 (a) Solve the following LPP using Big M method

[07]

$$\text{Maximize } z = 6x_1 + 4x_2$$

$$\text{subject to } x_1 + x_2 \leq 10, 2x_1 + x_2 \geq 4, x_1 \geq 0, x_2 \geq 0$$

- (b) Solve the following problem using Dual-Simplex Method.

[07]

$$\text{Maximize } Z = -3x_1 - x_2$$

$$\text{Subject to, } x_1 + x_2 \geq 1$$

$$2x_1 + 3x_2 \geq 2$$

$$x_1, x_2 \geq 0$$

- Q 3 Explain the following terms with reference to Queuing System

- (a) (i) Queue discipline (ii) The symbol a/b/c: d / f for queuing model

[04]

- (b) For a simple Queue system, if the traffic intensity is 0.5, 0.8, 0.9 and 0.95 respectively,

[05]

(i) What is the average number of customers in the system?

(ii) If the mean service rate is 10 per hour, what is the average time a customer is in the queue

- (c) Discuss the advantages and disadvantages of simulation

[05]

OR

- Q 3 (a) At a telephone booth assume that the mean duration of telephone conversation is three minutes, and that no more than a three – minute (average) wait for the phone may be tolerated; Determine the maximum call rate that can be supported by one telephone booth.

[03]

- (b) For M/M/1: ∞ /FIFO queuing system write the expressions for

[03]

(i) Probability that the system is empty

(ii) Average number of units in the system

(iii) Average waiting time in system

- (c) Observations of past data show the following patterns in respect of inter – arrival durations and service durations in a single queuing system. Using the random number table below simulate the queue behavior for a period of 30 minutes. [08]
- Random Nos: 93, 14, 72, 10, 21, 81, 87, 90, 38, 10

Inter Arrival Time

Service Time

Minutes	2	4	6	8	10
Probability	0.15	0.23	0.35	0.17	0.1

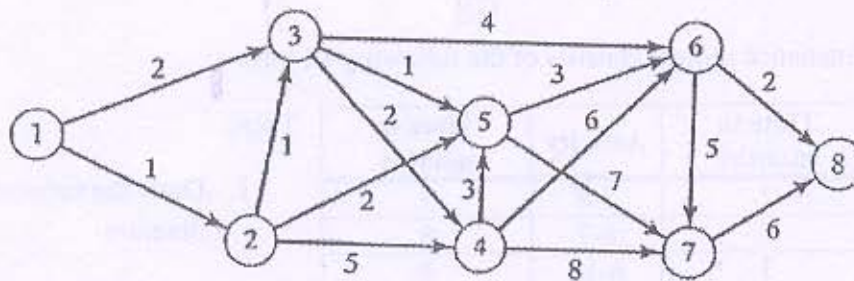
Minutes	1	3	5	7	9
Probability	0.1	0.22	0.35	0.23	0.1

Construct the simulation table:

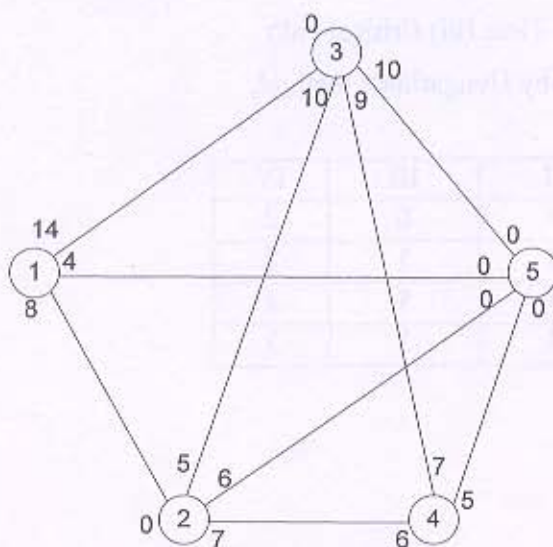
Random Number	Inter Arrival Time (min)	Arrival Time	Service Starts (min)	Random Number	Service Time (min)	Service Ends (min)	Waiting Time		
							Attendant (min)	Customer (min)	Queue Length

SECTION – II

- Q 4 The network given in below figure gives the distances in miles between pairs of cities 1,2,...,8. Use Dijkstra's Algorithm and find the shortest route between 1 and every other node in the given network. [07]



- Q 5 (a) Determine the maximal flow and the optimum flow in each arc for the network in following figure [07]



(b) A project has following characteristics.

[07]

Activity	Most optimistic time(a)	Most pessimistic time(b)	Most likely time(m)
1-2	1	5	1.5
2-3	1	3	2
2-4	1	5	3
3-5	3	5	4
4-5	2	4	3
4-6	3	7	5
5-7	4	6	5
6-7	6	8	7
7-8	2	6	4
7-9	5	8	6
8-10	1	3	2
9-10	3	7	5

Construct a pert network. Find critical path and variance for each event.

OR

Q 5 (a) A small maintenance project consists of the following 12 jobs.

[08]

Activity	Time in months	Activity	Time in months
1-2	2	5-8	5
2-3	7	6-7	8
2-4	3	6-10	4
3-4	3	7-9	4
3-5	5	8-9	1
4-6	3	9-10	7

Then,

1. Draw the network diagram
2. Find the total float, free float and independent float
3. Obtain the critical path

(b) Explain the following terms with reference to Project Network

[06]

(i) Earliest Start Time (ii) Earliest Finish Time (iii) Critical Path

Q 6 (a) Solve the following assignment problem by Hungarian's method.

[07]

	I	II	III	IV
A	8	7	0	2
B	6	4	3	2
C	6	5	5	4
D	7	5	3	3

(b) Obtain Initial Basic Feasible Solution of the following TP by VAM.

[07]

	D ₁	D ₂	D ₃	D ₄	Supply
O ₁	5	3	6	2	19
O ₂	4	7	9	1	37
O ₃	3	4	7	5	34
Demand	16	18	31	25	

OR

Q6(a) What is unbalanced transportation problem? How to resolve it?

[03]

(b) Find the Initial Basic Feasible Solution of the following TP by Least cost method.

[04]

	D ₁	D ₂	D ₃	D ₄	Supply
O ₁	6	3	5	4	22
O ₂	5	9	2	7	15
O ₃	5	7	8	6	08
Demand	07	12	17	09	

(c) Obtain an optimal basic feasible solution to the following TP.

[07]

		Warehouse				Capacity
		W1	W2	W3	W4	
Factory	F1	19	30	50	10	7
	F2	70	30	40	60	9
	F3	40	8	70	20	18
Requirement		5	8	7	14	